



Modeling Dark Matter Stars

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Introduction

What is Dark Matter?

Dark Matter is a type of matter that interacts weakly with other matter and is, therefore, difficult to detect with current observations and technologies.

How do we know that Dark Matter exists, and what is its purpose?

We know that Dark Matter exists because it explains the observed discrepancies within the velocities of galaxies and galaxy clusters. The magnitude of these velocities is much greater than what they should be, based on the observable mass they contain. Therefore, it implies that there are significant amounts of Dark Matter within galaxies and galaxy clusters.



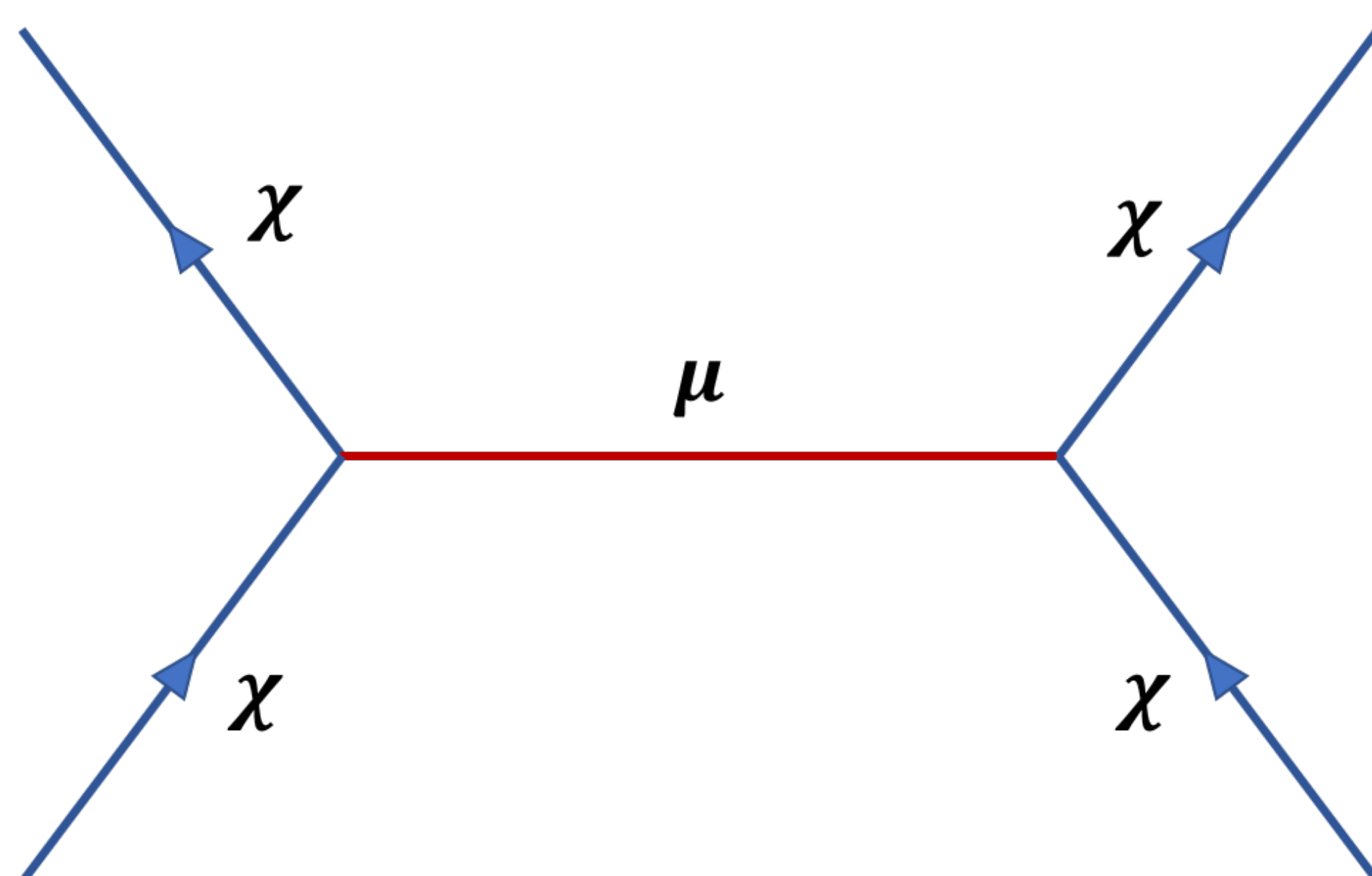
NASA, ESA, and D. Coe (JPL/Caltech and STScI)

To the left is an image of galaxy cluster Abell 1698, which shows a distribution of Dark Matter concentrations produced by researchers at NASA. The tints of blue are the inferred Dark Matter concentrations within the galaxy cluster.

Dark Matter Interactions

Interactions between Dark Matter particles can depend on multiple variables

- Dark Matter Particle (χ) and its mass (m_χ)
- Field Mediator (μ) and its mass (m_μ)
- Interaction Coupling Strength (α)



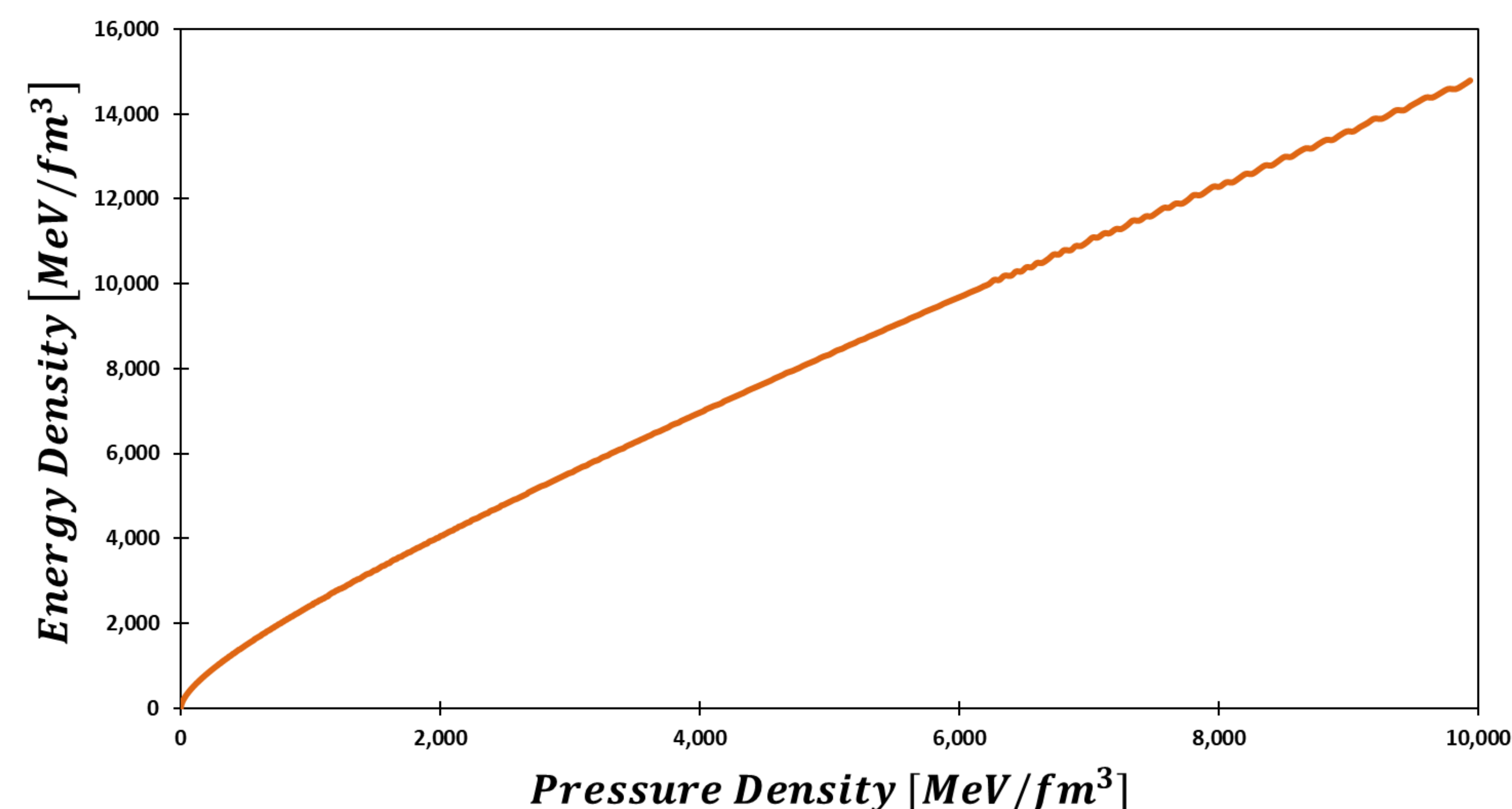
Modeling Dark Matter Stars

Modeling Dark Matter Stars was computationally done using Python, the NumPy library, and the following values.

- $m_\chi = 1000 \text{ MeV}/c^2$
- $m_\mu = 10 \text{ MeV}/c^2$
- $\alpha = 10^{-3}$

The Equation of State was calculated using the construction provided in "Asymmetric dark matter stars", resulting in corresponding pressure densities and energy densities for Dark Matter Stars (Kouvaris et al., 2015).

Energy Density vs Pressure Density



Afterward, the following system of differential equations was numerically solved and used to calculate a total mass and radius of a Dark Matter Star, given a central pressure density and energy density.

Tolman-Oppenheimer-Volkoff Equation(s)

$$\frac{dP}{dr} = - \left(\frac{Gm(r)\rho(r)}{r^2 c^2} \right) \left(1 + \frac{P(r)}{\rho(r)} \right) \left(1 + \frac{4\pi r^3 P(r)}{m(r)} \right) \left(1 - \frac{2Gm(r)}{rc^2} \right)^{-1}$$

$$\frac{dM}{dr} = 4\pi r^2 \rho(r)$$

The following properties are dependent on a given radius within a Dark Matter Star:

- m – mass
- ρ – energy density
- P – pressure density

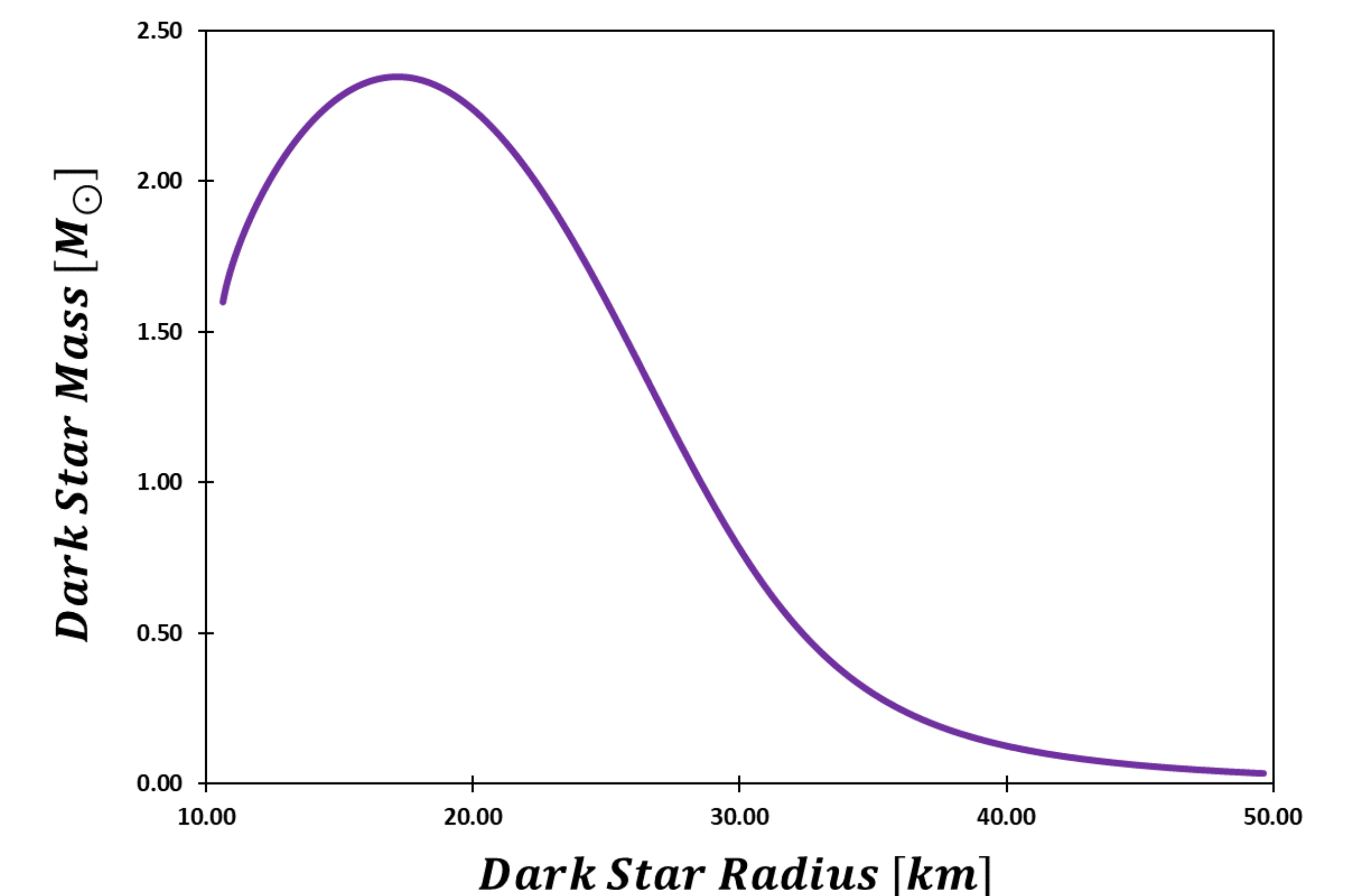
Resulting Dark Matter Stars

The Tolman-Oppenheimer-Volkoff equations were used to calculate the total masses and radii for a family of Dark Matter Stars.

Within the graph, the total masses are expressed as multiples of solar masses.

Solar Mass ($M_\odot = 1.989 \times 10^{30} \text{ kg}$)

Dark Star Mass vs Dark Star Radius



Based on these results, the largest possible Dark Matter Star would be approximately 2.347 solar masses with a radius of 17.197 kilometers.

References

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https://www.nasa.gov/mission_pages/hubble/science/dark-matter-map.html
- Kouvaris, Chris and Niklas Grønlund Nielsen. "Asymmetric dark matter stars." Physical Review D, vol. 92, no. 6, 2015.
- Maselli, Andrea, et al. "Dark stars: Gravitational and electromagnetic observables." Physical Review D, vol. 96, no. 2, 2017.
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